A FLAT GROUP SCHEME OF FINITE TYPE OVER INFINITE-DIMENSIONAL AFFINE SPACE WHICH IS NOT QUASI-SEPARATED

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is the A^infty instantiation of the context construction; the author's 'We omit the verification that this defines a group scheme structure' sits inside the context span (construction layer), not the counted proof; content verbatim.

Editorial note. The label lemma-non-quasi-separated-group-scheme names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3830-3860: the doubled-origin construction over the affine line - G = affine line with 0 doubled (Schemes, example-affine-space-zero-doubled) over S = A^1_k, fibres singletons or two-point sets with the U-point as zero, two opens U, V mapping isomorphically to S with U cap V -> S \ {0}, the multiplication m : G x_S G -> G defined piecewise on the four decomposition pieces, with the author's own omitted verification of the group scheme structure
- Schemes, example-not-quasi-separated: the doubled-origin line is not quasi-separated; base-change behaviour of quasi-separatedness

Source reference. Original lemma 047L: sources/stacks-chow/tags.txt, source lines 5512–5512. Prerequisite pointer invoked at 3887; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 01JD: sources/stacks-chow/tags.txt, source lines 2010–2010. Prerequisite pointer invoked at 3893; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 01KL: sources/stacks-chow/tags.txt, source lines 2054–2054. Prerequisite pointer invoked at 3881; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Every group scheme over a field is separated, see Groupoids, Lemma 047L. This is not true for group schemes over a base.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let k be a field. Let $S = \mathrm{Spec}(k[x]) = \mathbf{A}^1_k$. Let G be the affine line with 0 doubled (see Schemes, Example 01JD) seen as a scheme over S. Thus a fibre of $G \to S$ is either a singleton or a set with two elements (one in U and one in V). Thus we can endow these fibres with the structure of a group (by letting the element in U be the zero of the group structure). More precisely, G has two opens U, V which map isomorphically to S such that $U \cap V$ is mapped isomorphically to $S \setminus \{0\}$. Then

$$G \times_S G = U \times_S U \cup V \times_S U \cup U \times_S V \cup V \times_S V$$

where each piece is isomorphic to S. Hence we can define a multiplication $m : G \times_S G \to G$ as the unique S-morphism which maps the first and the last piece into U and the two middle pieces into V. This matches the pointwise description given above. We omit the verification that this defines a group scheme structure.

Lemma 0.1. *There exists a flat group scheme of finite type over the infinite dimensional affine space which is not quasi-separated.*

Proof. The same construction as above can be carried out with the infinite dimensional affine space $S = \mathbf{A}^\infty_k = \mathrm{Spec}\, k[x_1, x_2, \ldots]$ as the base and the origin $0 \in S$ corresponding to the maximal ideal $(x_1, x_2, \ldots)$ as the closed point which is doubled in G. The resulting group scheme $G \to S$ is not quasi-separated as explained in Schemes, Example 01KL. □